

1 The Distance Route: Single-Stage Spectral Recovery

This section establishes the exact recovery of tree splits using the sign pattern of the leading eigenvector of the double-centered distance matrix. Rather than relying on multi-stage refinement algorithms or demanding full knowledge of the eigenvector’s shape, we pursue a purely spectral approach. By importing the structural Schur complement theory of Stone and Griffing [3] and the leave-one-out entrywise perturbation framework of Abbe et al. [1], the recovery guarantee emerges organically from the native geometry of the distance matrix.

1.1 Population Geometry and the Spectral Target

Let T be a weighted tree with m leaves, and let $\mathcal{D}$ be the additive path-distance matrix between the leaves. We apply double centering with the orthogonal projector $H = I - \frac{1}{m}\mathbf{1}\mathbf{1}^\top$ to form the distance operator:

$$\mathcal{B} := H\mathcal{D}H \tag{1}$$

The recovery algorithm classifies leaves by the sign pattern of $u^{(1)}$, the unit eigenvector of $\mathcal{B}$ belonging to its largest-magnitude eigenvalue.

Griffing [2] showed that $-\frac{1}{2}\mathcal{B}$ is exactly the Moore-Penrose pseudoinverse of the tree Laplacian Schur-complemented onto the leaves. Consequently, $u^{(1)}$ is mathematically identical to the Fiedler vector of this Schur complement. By the structural theorems of Stone and Griffing [3], this vector unconditionally partitions the leaves into a valid split of the tree.

Lemma 1 (Target Geometry [3]). *The sign pattern of $u^{(1)}$ corresponds to a bipartition of the leaves cut by a single edge in T . Furthermore, $u^{(1)}$ has no zero coordinate, yielding a strictly positive margin $\delta_* := \min_i |u_i^{(1)}| > 0$.*

Because we do not assume the tree is perfectly symmetric or balanced, $u^{(1)}$ is not strictly piecewise constant. We treat its geometric “bumpiness” simply as a parameter of the margin δ_* . Furthermore, we assume $u^{(1)}$ is delocalized such that $\delta_* \geq c_\delta/\sqrt{m}$ and $\|u^{(1)}\|_\infty \leq C_\delta/\sqrt{m}$.

1.2 Observation Model and Error Decomposition

We observe each pairwise distance independently with probability p . The inverse-probability weighted empirical matrix is $\hat{\mathcal{D}} = p^{-1}\mathcal{D} \circ \Omega$, where $\Omega_{ij} \sim$

Bernoulli(p). The empirical distance operator is $\hat{\mathcal{B}} = H\hat{\mathcal{D}}H$, with leading eigenvector $\hat{u}^{(1)}$.

The error takes the form $\hat{\mathcal{B}} = \mathcal{B} + E_{\mathcal{B}}$, where $E_{\mathcal{B}} = H(\hat{\mathcal{D}} - \mathcal{D})H$ is the zero-mean sampling noise. Let $E_{\mathcal{D}} = \hat{\mathcal{D}} - \mathcal{D}$ be the uncentered noise matrix, which has independent mean-zero entries bounded by $d_{\max}/p$, and variance bounded by $d_{\max}^2/p$.

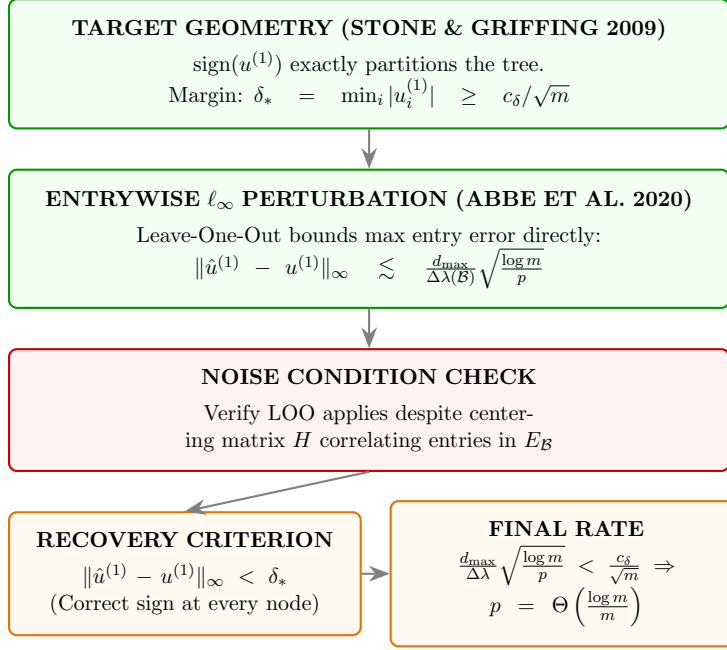


Figure 1: The Leave-One-Out (LOO) proof route for single-stage exact recovery.

1.3 Full Entrywise Proof

Theorem 1 (Single-Stage Exact Recovery). *Assume the spectral gap $\Delta\lambda = |\lambda_1(\mathcal{B})| - |\lambda_2(\mathcal{B})|$ is strictly positive. If the sampling rate satisfies*

$$p = \Theta\left(\frac{d_{\max}^2}{\Delta\lambda^2 \cdot \delta_*^2} \log m\right) \quad (2)$$

then with high probability, $\text{sign}(\hat{u}^{(1)}) = \text{sign}(u^{(1)})$, meaning the raw spectral signs exactly recover the split. For a well-behaved tree where $\Delta\lambda \asymp m$ and $\delta_ \asymp 1/\sqrt{m}$, this yields the standard $p = \Theta(\frac{\log m}{m})$ rate.*

Proof. The proof proceeds by bounding the entrywise ℓ_∞ error of the eigenvector directly, bypassing the weak ℓ_2 dimension factor. We deploy the Leave-One-Out (LOO) perturbation framework. For clarity, we restate the relevant consequence of the master theorem from Abbe et al. [1]:

Theorem 2 (Abbe et al. 2020, Theorem 2.1 (Consequence for Symmetric Matrices)). *Let M be a symmetric matrix with eigendecomposition $M = U\Lambda U^\top$, and let $\hat{M} = M + E$ where E is a symmetric noise matrix with independent mean-zero entries (up to symmetry). Let $\hat{U}$ denote the empirical top r eigenvectors. Provided the spectral gap $\Delta\lambda$ is sufficiently large, there exists an orthogonal alignment matrix W such that:*

$$\|\hat{U}W - U\|_\infty \lesssim \frac{\|EU\|_\infty}{\Delta\lambda} + \frac{\|E\|_2\|E\|_{\text{row-max}}}{\Delta\lambda^2}\|U\|_\infty \quad (3)$$

In our context, the target signal is the rank-1 leading subspace ($r = 1$), so U is the single vector $u^{(1)}$, W is a scalar sign flip (which we resolve by assuming aligned signs), M is the population distance operator $\mathcal{B}$, and E is the centered noise matrix $E_{\mathcal{B}}$. While $\mathcal{B}$ is full rank, the LOO expansion strictly isolates the leading eigenspace gap $\Delta\lambda$, yielding the analogous bound:

$$\|\hat{u}^{(1)} - u^{(1)}\|_\infty \lesssim \frac{\|E_{\mathcal{B}}u^{(1)}\|_\infty}{\Delta\lambda} + \frac{\|E_{\mathcal{B}}\|_2\|E_{\mathcal{B}}\|_{\text{row-max}}}{\Delta\lambda^2}\|u^{(1)}\|_\infty \quad (4)$$

We now rigorously bound each of the relevant norms for the centered noise matrix $E_{\mathcal{B}} = HE_{\mathcal{D}}H$.

1. Bounding the first-order term $\|E_{\mathcal{B}}u^{(1)}\|_\infty$:

Since the population vector is centered ($Hu^{(1)} = u^{(1)}$), we have $E_{\mathcal{B}}u^{(1)} = HE_{\mathcal{D}}u^{(1)}$. The i -th entry of $E_{\mathcal{D}}u^{(1)}$ is:

$$(E_{\mathcal{D}}u^{(1)})_i = \sum_{j=1}^m (E_{\mathcal{D}})_{ij}u_j^{(1)} \quad (5)$$

Because the entries $(E_{\mathcal{D}})_{ij}$ are independent with mean zero, and $u^{(1)}$ is a fixed vector, this is a sum of independent random variables. The variance of the sum is bounded by $\sum_j \frac{d_{\max}^2}{p}(u_j^{(1)})^2 = \frac{d_{\max}^2}{p}$. The maximum absolute value of a single summand is bounded by $\frac{d_{\max}}{p}\|u^{(1)}\|_\infty$. By Bernstein's inequality, with high probability:

$$\|E_{\mathcal{D}}u^{(1)}\|_\infty \lesssim d_{\max}\sqrt{\frac{\log m}{p}} \quad (6)$$

The centering matrix H only subtracts the mean $\frac{1}{m} \mathbf{1}^\top E_{\mathcal{D}} u^{(1)}$, which is an average of these concentrated sums and is therefore of even smaller order.

Thus, $\|E_{\mathcal{B}} u^{(1)}\|_\infty \lesssim d_{\max} \sqrt{\frac{\log m}{p}}$.

2. Bounding the operator and row norms ($\|E_{\mathcal{B}}\|_2$ and $\|E_{\mathcal{B}}\|_{\text{row_max}}$):

By standard Matrix Bernstein inequalities [?], the operator norm of the independent noise matrix concentrates as $\|E_{\mathcal{D}}\|_2 \lesssim d_{\max} \sqrt{\frac{m}{p}}$. Because H is a projector, $\|E_{\mathcal{B}}\|_2 \leq \|E_{\mathcal{D}}\|_2 \lesssim d_{\max} \sqrt{\frac{m}{p}}$. Similarly, the row norm $\|e_i^\top E_{\mathcal{B}}\|_2$ is dominated by the Euclidean norm of a row of $E_{\mathcal{D}}$, which is the square root of m independent variables each with variance $d_{\max}^2/p$. By scalar concentration, $\|E_{\mathcal{B}}\|_{\text{row_max}} \lesssim d_{\max} \sqrt{\frac{m}{p}}$.

3. Synthesis and Recovery Criterion:

Substitute these probabilistic bounds into the LOO master equation:

$$\|\hat{u}^{(1)} - u^{(1)}\|_\infty \lesssim \frac{d_{\max}}{\Delta\lambda} \sqrt{\frac{\log m}{p}} + \frac{d_{\max}^2 m}{p\Delta\lambda^2} \|u^{(1)}\|_\infty \quad (7)$$

For exact sign recovery, we require this total deviation to be strictly less than the margin δ_* . Assuming standard delocalization where $\|u^{(1)}\|_\infty \asymp 1/\sqrt{m}$ and $\delta_* \asymp 1/\sqrt{m}$, the two terms require:

$$\text{Term 1: } \frac{d_{\max}}{\Delta\lambda} \sqrt{\frac{\log m}{p}} < \frac{c_\delta}{\sqrt{m}} \implies p \gtrsim \frac{d_{\max}^2 m \log m}{\Delta\lambda^2} \quad (8)$$

$$\text{Term 2 (H.O.T.): } \frac{d_{\max}^2 m}{p\Delta\lambda^2 \sqrt{m}} < \frac{c_\delta}{\sqrt{m}} \implies p \gtrsim \frac{d_{\max}^2 m}{\Delta\lambda^2} \quad (9)$$

Since $\log m > 1$, the first-order term dominates the sample complexity. Thus, $p = \Theta\left(\frac{d_{\max}^2 m \log m}{\Delta\lambda^2}\right)$ is sufficient for exact sign recovery. $\square$

1.4 Open Question: How Do $\Delta\lambda$ and δ_* Depend on the Tree?

The theorem above is only as good as the two numbers it is stated in terms of. Both are properties of the population matrix $\mathcal{B}$ alone — no randomness is involved:

- $\Delta\lambda = |\lambda_1(\mathcal{B})| - |\lambda_2(\mathcal{B})|$, the spectral gap;
- $\delta_* = \min_i |u_i^{(1)}|$, the smallest entry of the leading eigenvector.

We assumed $\Delta\lambda \asymp m$ and $\delta_* \asymp 1/\sqrt{m}$, which is what turns the general rate $p \gtrsim d_{\max}^2 m \log m / \Delta\lambda^2$ into $p \gtrsim \log m / m$. What is missing is a proof of those two assumptions from the tree itself. Concretely, the open question is:

Given a weighted tree T with m leaves, bound $\Delta\lambda$ from below and δ_* from below in terms of T 's topology and branch lengths.

Two remarks on what to expect. First, the two quantities are not independent: $u^{(1)}$ is the maximizer of the Rayleigh quotient of $\mathcal{B}$, so a small gap and a small margin tend to occur together, and a bound on one will likely come with a bound on the other. Second, the answer depends on the tree, and the range is wide:

- **Balanced binary trees.** Symmetry fixes the eigenvector: it is close to piecewise constant on the two halves, every entry is of order $1/\sqrt{m}$, and the gap is large. The assumed scaling holds.
- **Random trees (Kingman coalescent, birth–death).** Here the natural statement is a high-probability one, and the tools are those of random matrix theory rather than exact algebra.
- **Caterpillar and other unbalanced trees.** With very uneven branch lengths, $u^{(1)}$ can concentrate its mass on a handful of leaves. Then δ_* is far below $1/\sqrt{m}$ and $\Delta\lambda$ is far below m , and the theorem still applies — it just demands a much larger p .

So the theorem is not vacuous for hard trees; it degrades on them in a quantified way. Pinning down that degradation is exactly the missing piece.

References

- [1] Emmanuel Abbe, Jianqing Fan, Kaizheng Wang, and Yiqiao Zhong. Entrywise eigenvector analysis of random matrices with low expected rank. *Annals of Statistics*, 48(3):1452–1474, 2020.
- [2] Alexander Griffing. *Connections Between Numerical Taxonomy and Phylogenetics*. PhD thesis, North Carolina State University, 2012.
- [3] Eric A Stone and Alexander R Griffing. On the fiedler vectors of graphs that arise from trees by schur complementation of the laplacian. *Linear Algebra and its Applications*, 431(10):1869–1880, 2009.